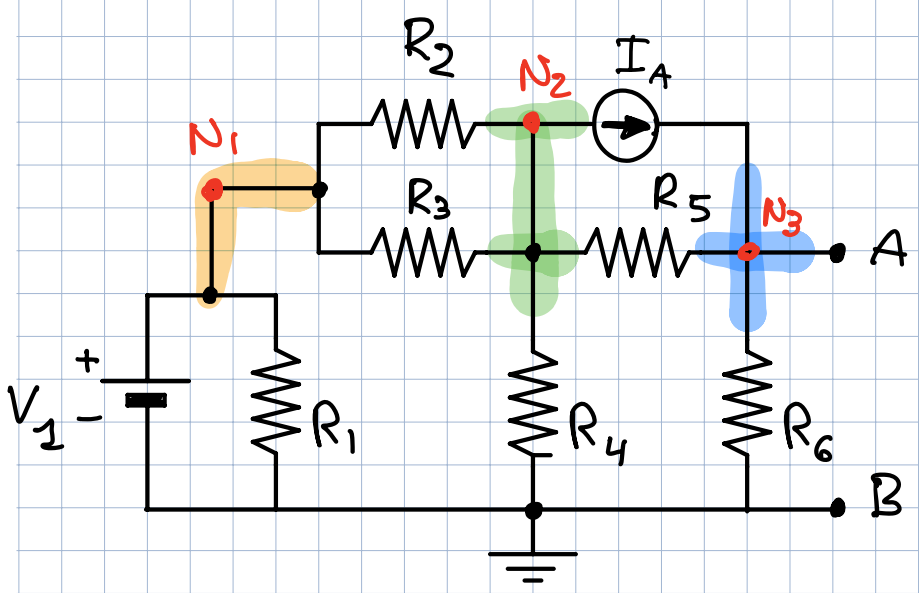
# ESERCIZIO 3




DATI

$R_1 = 10\,K\Omega$

$R_2 = 4\,K\Omega$

$R_3 = 4\,K\Omega$

$R_4 = 2\,K\Omega$

$R_5 = 1\,K\Omega$

$R_6 = 2\,K\Omega$

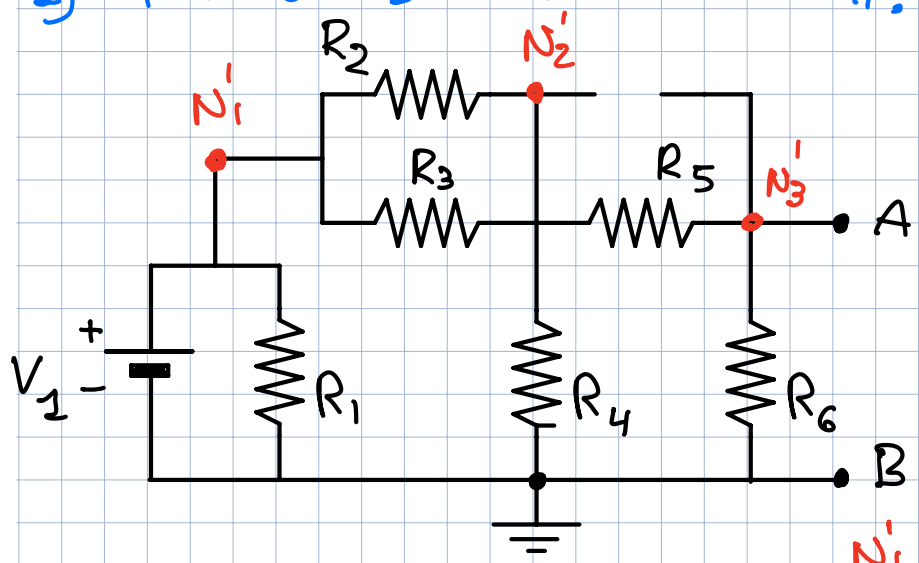
$I_A = 10\,mA$

$V_1 = 10\,V$

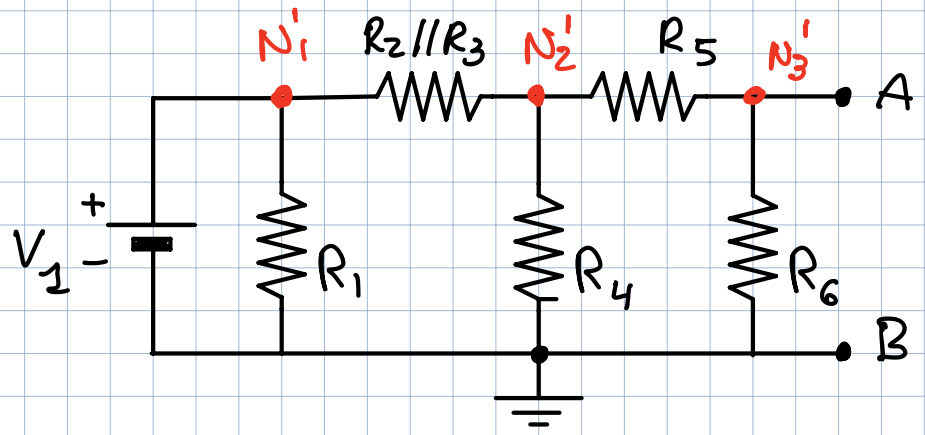
PER QUESTO ESERCIZIO CALCOLARE IL GENERATORE EQUIVALENTE DI THEVENIN E NORTON VISTO DAI MORSETTI A-B

RISOLVO IL CIRCUITO USANDO LA SOVRAPPOSIZIONE DEGLI EFFETTI

1) ATTIVO $V_1$ e SPENGO $I_A$:

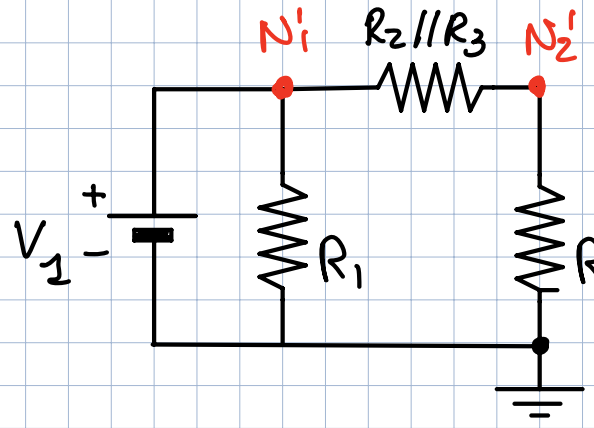



EQUIVALENTE




$V'_{N_1} = V_1$

$V'_{N_2}$ ?




$R_2 // R_3 = 4K\Omega // 4K\Omega = 2K\Omega$

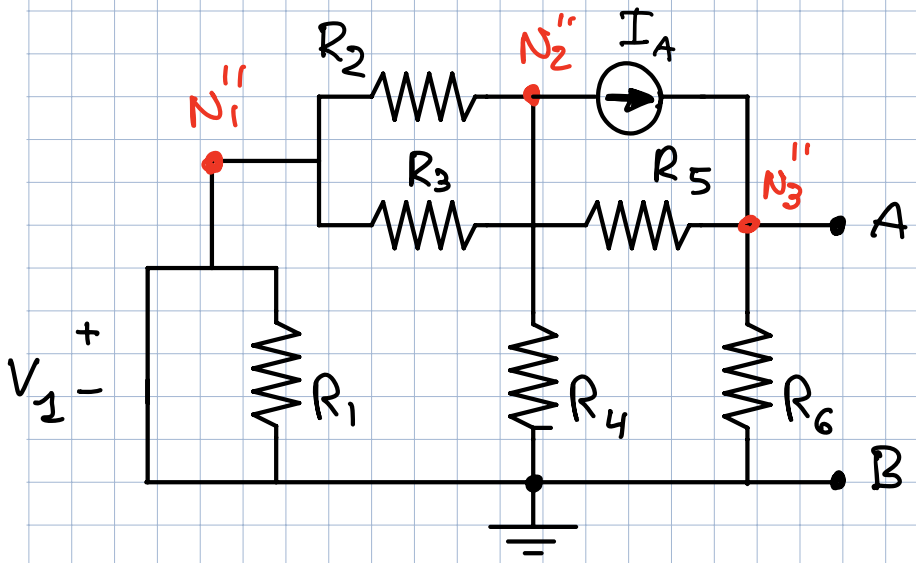
$R_4 // (R_5 + R_6) = 2K\Omega // 3K\Omega = \frac{6}{5} K\Omega = 1{,}2 K\Omega$

$$V'_{N_2} = V_1 \cdot \frac{R_4 // (R_5 + R_6)}{R_2 // R_3 + R_4 // (R_5 + R_6)} = V_1 \cdot \frac{1{,}2\ K\Omega}{3{,}2\ K\Omega} = 3{,}75\,V$$

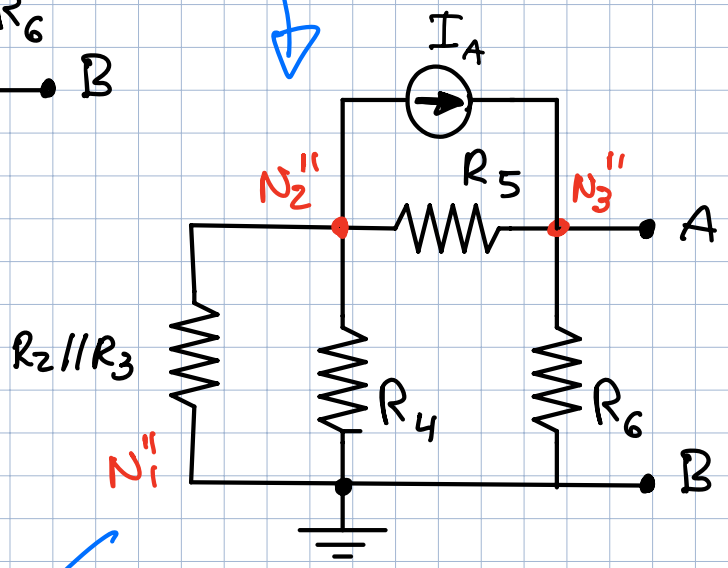
PER CALCOLARE $V'_{N_3}$, USO IL CIRCUITO INIZIALE:

$$V'_{N_3} = V'_{N_2} \cdot \frac{R_6}{R_5 + R_6} = V'_{N_2} \cdot \frac{2}{3} K\Omega = 2{,}5V$$

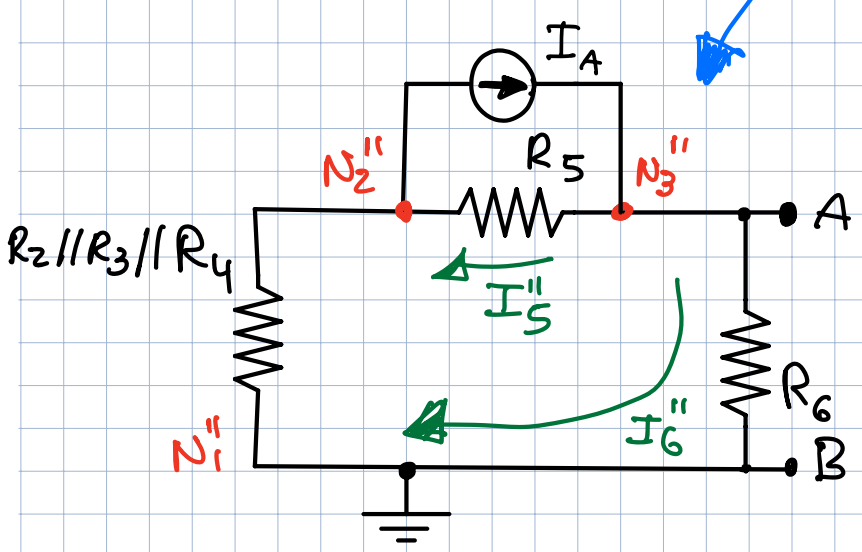
2) ATTIVO $I_A$ e SPENGO $V_1$:




EQUIVALENTE




EQUIVALENTE




PARTITORE DI CORRENTE

$R_2 // R_3 // R_4 =$

$= 4KR // 4KR // 2KR = 1KR$

$$I_5'' = I_A \cdot \frac{R_6 + R_2 // R_3 // R_4}{R_5 + R_6 + R_2 // R_3 // R_4} = \frac{2KR + 1KR}{1KR + 2KR + 1KR} I_A$$

$$= \frac{3}{4} I_A = 7{,}5 mA$$

$$I_6'' = I_A \cdot \frac{R_5}{R_5 + R_6 + R_2 // R_3 // R_4} = \frac{1KR}{4KR} I_A = 2{,}5 mA$$

$\Rightarrow V_{N_1}'' = 0V$

$V_{N_2}'' = - I_6'' R_2 // R_3 // R_4 = -2{,}5 mA \times 1KR = -2{,}5V$

$V_{N_3}'' = I_6'' R_6 = 2{,}5 mA \cdot 2KR = 5V$

$\Rightarrow V_{N_1} = V_{N_1}' + V_{N_1}'' = 10V$

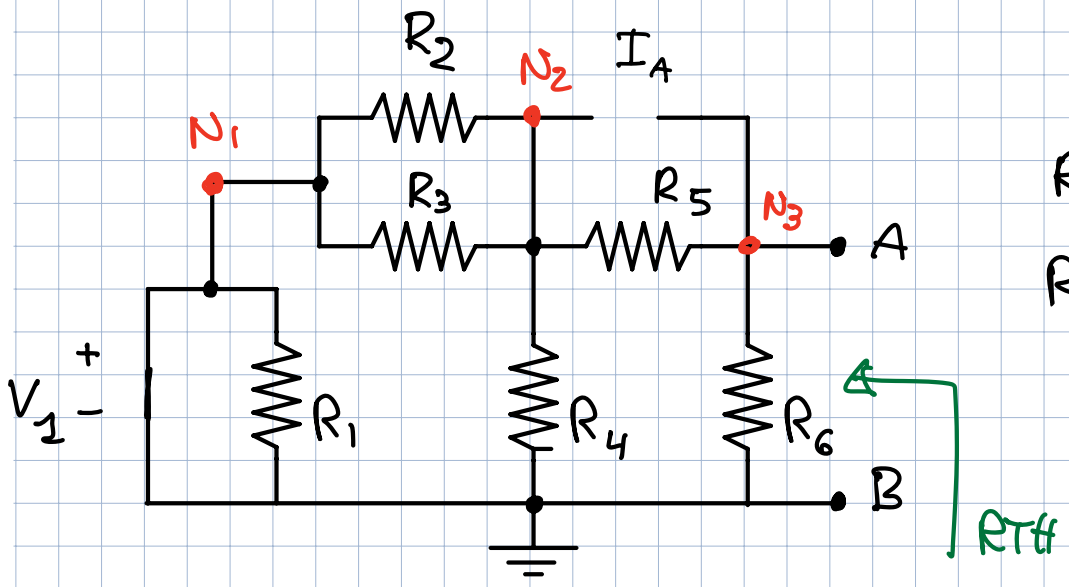
$V_{N_2} = V_{N_2}' + V_{N_2}'' = 1{,}25V$

$V_{N_3} = V_{N_3}' + V_{N_3}'' = 7{,}5V \Rightarrow V_{th} = V_{AB} = 7{,}5V$

CON QUESTE TENSIONI AI NODI POSSO TROVARE LE CORRENTI SU TUTTO IL CIRCUITO

TROVO LA $R_{TH}$:

ANNULLO GENERATORI INDIPENDENTI

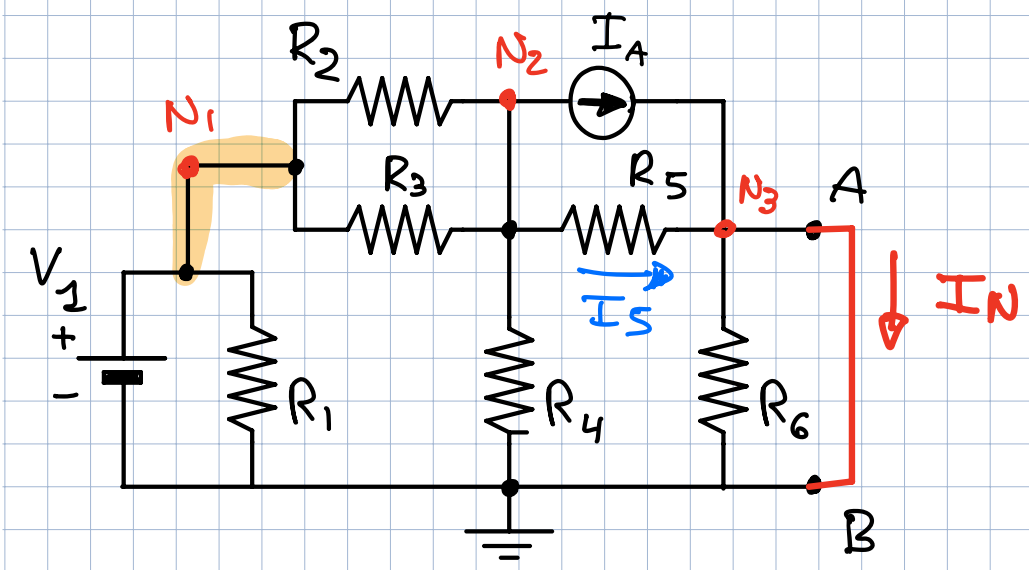



$R_1$ è CORTOCIRCUITATA

$R_1$, $R_3$ e $R_4$ SONO IN PARALLELO

$$\Rightarrow R_{TH} = R_6 \,//\, [R_5 + R_1 // R_2 // R_3] = 2k\Omega \,//\, [1k\Omega + 1k\Omega] = 1k\Omega$$

EQUIVALENTE DI NORTON




NOTO CHE:

$R_6$ CORTOCIRCUITATA

$\Rightarrow I_{R_6} = 0\,A$

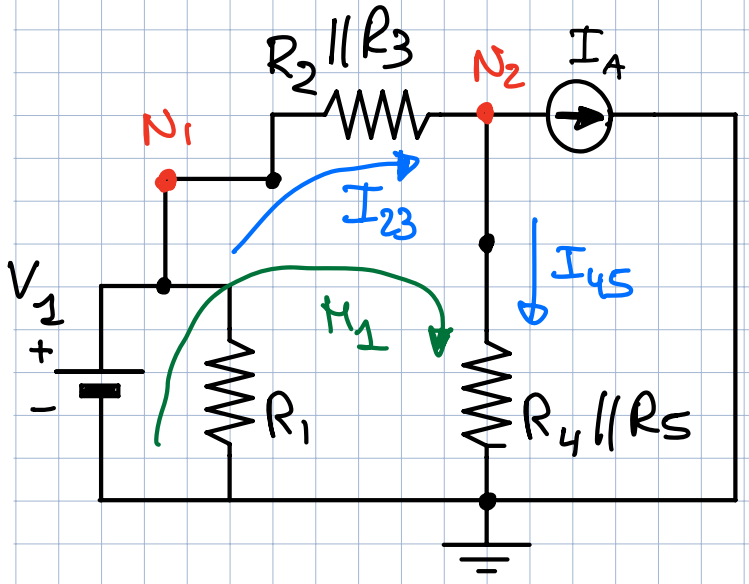
$R_4$ e $R_5$ SONO IN PARALLELO

$V_{N_3} = 0$

LA CORRENTE $I_N = I_A + I_5$

$$I_5 = \frac{V_{N_2} - V_{N_3}}{R_5} = \frac{V_{N_2}}{R_5}$$

CALCOLO $V_{N_2}$




N.B. IN QUESTO CIRCUITO EQUIVALENTE X IL CALCOLO DI $V_{N_2}$, IL RAMO DI $I_N$ È "SCOMPARSO"!

VA BENE VISTO CHE DEVO CALCOLARE $V_{N_2}$!

MAGLIA $M_1$:
$$\begin{cases} -V_1 + I_{23} R_2 \| R_3 + I_{45} R_4 \| R_5 = 0 \\ I_{45} = I_{23} - I_A \end{cases}$$

$$V_1 = I_{23} \underbrace{R_2 \| R_3}_{= 2K\Omega} + I_{23} \underbrace{R_4 \| R_5}_{} - I_A \underbrace{R_4 \| R_5}_{0{,}667\,K\Omega}$$

$$\Rightarrow I_{23} = \frac{V_1 + I_A \cdot R_4 \| R_5}{R_2 \| R_3 + R_4 \| R_5} = \frac{10V + 6{,}67V}{2{,}667\,K\Omega} = 6{,}25\,mA$$

$$\Rightarrow I_{45} = I_{23} - I_A = -3{,}75\,mA$$

$$\Rightarrow V_{N_2} = I_{45} \cdot R_4 \| R_5 = -2{,}5V$$

QUINDI DAL CIRCUITO INI RIACE TROVO:

$$I_5 = \frac{V_{N_2}}{R_5} = -2{,}5\,mA$$

$$\Rightarrow I_N = I_A + I_5 = 7{,}5\,mA$$

VERIFICHIAMO CHE $R_{TH} = \frac{V_{TH}}{I_N}$ ?

$$R_{TH} = \frac{7{,}5V}{7{,}5mA} = 1\,K\Omega$$

C.V.D.